

Employing Metaprogramming: A Macro-Driven Strategy to Overcome Compiler-Imposed Limitations on Specialization

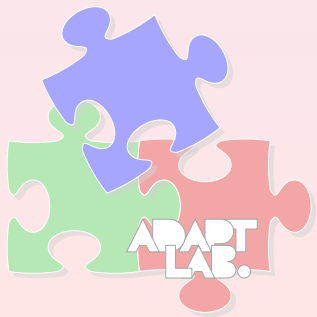
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L-31 - Computer Science
Academic Year 2024-2025





Specialization
State of the Art and Limitations

Specialization

Problem Statement

Proposal

By Example: Implementing the Proposal

Limitations and Future Work

Implementation Details

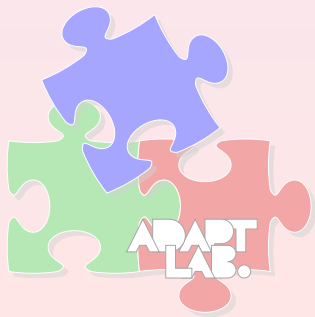
Evaluation

Conclusions

Systems programming aims to combine generic programming with type-specific optimizations.

Mechanisms vary widely, from mature and widely used systems to experimental features, with varying degrees of success.

In many modern ecosystems, specialization is often unstable, forcing a tradeoff between performance and compiler stability.



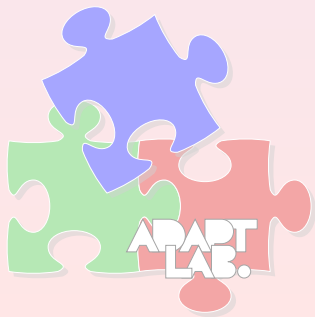


Problem Statement
What's happening in Rust?

In 2015, an RFC introduced specialization to Rust as an unstable feature confined to the Nightly Channel.

However, due to unresolved design issues and concerns about long-term maintenance, specialization is still unstable.

Without stable specialization, developers often resort to code duplication or complex workarounds.



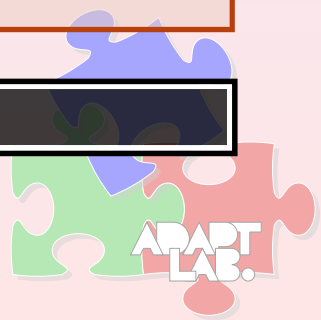


Problem Statement
Motivational Example

Consider implementing a trait for different types, with a specialized implementation for a specific type.

```
1  trait MyTrait {
2      fn my_method(&self);
3  }
4  impl<T> MyTrait for T {
5      fn my_method(&self) {
6          println!("Generic implementation");
7      }
8  }
9  impl MyTrait for i32 {
10     fn my_method(&self) {
11         println!("Specialized for i32");
12     }
13 }
14 fn main() {
15     10.my_method();   // should print "Specialized for i32"
16     10.5.my_method(); // should print "Generic implementation"
17 }
```

```
error[E0119]: conflicting implementations of trait MyTrait for type i32
```





Proposal

Specialization

Problem Statement

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By Example: Implementing the Proposal

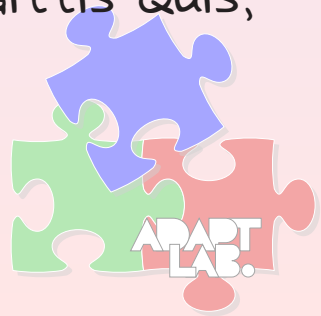
Limitations and Future Work

Implementation Details

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By Example: Implementing the Proposal

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Limitations and Future Work

Specialization

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Implementation Details
Conditions Definition

Specialization

Problem Statement

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Implementation Details
Method Wrapping

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Implementation Details
Build Scripts

Specialization

Problem Statement

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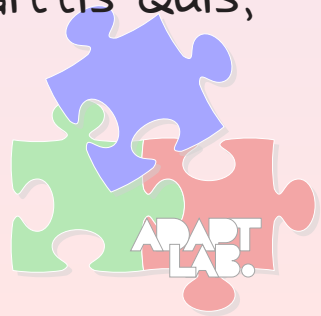
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Evaluation
Compiler Instrumentation

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Evaluation
Specialization Heuristics

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Evaluation
Results

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Conclusions

Interesting results:

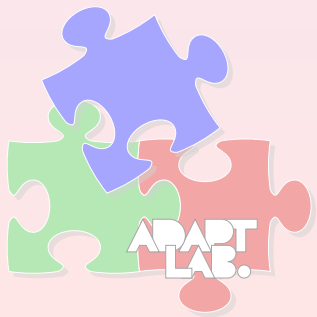
- We are writing an article (XXX) to be submitted to XXX.

Interesting twist:

- XXX

Future work:

- XXX





Thanks for your attention!

